

Problem Statement

Evaluate the limit from the image "dailyintegral-limit-limit-1x1-2026-08-06.jpg":

$$\lim_{n \rightarrow \infty} \sum_{x=2}^n (\zeta(x) - 1)$$

Detailed Solution

To evaluate this limit, we express the Riemann zeta function as an infinite series, which will allow us to convert the expression into a double sum. By swapping the order of summation, we can evaluate a geometric series and then a telescoping series.

Step 1: Expand the Riemann Zeta Function

The Riemann zeta function for $x > 1$ is defined as the infinite series:

$$\zeta(x) = \sum_{k=1}^{\infty} \frac{1}{k^x} = 1 + \frac{1}{2^x} + \frac{1}{3^x} + \dots$$

Therefore, subtracting 1 removes the first term of the series, leaving us with:

$$\zeta(x) - 1 = \sum_{k=2}^{\infty} \frac{1}{k^x}$$

Step 2: Set Up the Double Sum and Swap the Order

Substitute this series back into the original limit expression. As $n \rightarrow \infty$, our finite sum becomes an infinite series:

$$\lim_{n \rightarrow \infty} \sum_{x=2}^n (\zeta(x) - 1) = \sum_{x=2}^{\infty} \sum_{k=2}^{\infty} \frac{1}{k^x}$$

Since all terms in the double sequence $\frac{1}{k^x}$ are positive, Tonelli's Theorem allows us to interchange the order of summation without changing the result:

$$\sum_{k=2}^{\infty} \sum_{x=2}^{\infty} \frac{1}{k^x}$$

Step 3: Evaluate the Inner Geometric Series

Looking at the inner sum over x , we hold k constant. This forms an infinite geometric series with the first term $a = \frac{1}{k^2}$ (since x starts at 2) and the common ratio $r = \frac{1}{k}$:

$$\sum_{x=2}^{\infty} \left(\frac{1}{k}\right)^x = \frac{a}{1-r} = \frac{\frac{1}{k^2}}{1-\frac{1}{k}}$$

Multiply the numerator and denominator by k^2 to simplify:

$$\frac{1}{k(k-1)}$$

Using partial fraction decomposition, we can rewrite this expression as:

$$\frac{1}{k-1} - \frac{1}{k}$$

Step 4: Evaluate the Telescoping Series

Now, substitute this simplified result back into the outer sum over k :

$$\sum_{k=2}^{\infty} \left(\frac{1}{k-1} - \frac{1}{k} \right)$$

This is a classic telescoping series. Let's write out the first few terms to observe the cancellation:

$$\left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots$$

As $k \rightarrow \infty$, all intermediate terms cancel out perfectly, leaving only the very first term from the $k = 2$ calculation and the limit of the subtracted term, which goes to zero:

$$1 - \lim_{k \rightarrow \infty} \frac{1}{k} = 1 - 0 = 1$$

Final Answer:

1